

Exploring Pathways Associated with Loss of Swarming Motility in Uropathogenic *Escherichia coli*



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Introduction

Uropathogenic *Escherichia coli* (UPEC) is the most common cause of urinary tract infections (UTIs). Swarming is a virulence factor of UPEC, and is characterized by particular flagella-powered movement over solid media (Fig. 1a). *E. coli* CFT073 $\Delta toIC$ is a mutant of the uropathogenic model strain *E. coli* CFT073 that is able to swim but unable to swarm (Fig.1b). TolC is the main efflux pump of Gram-negative bacteria (Fig. 2), and its link with swarming is not understood. Ongoing results from our lab suggest that molecules secreted when UPEC CFT073 grows in swarming agar, such as surfactants or signaling factors, may be able to rescue the *toIC* phenotype.

In an effort to understand this phenotype, we looked at other gene deletions that result in non swarming. Based on this information, we propose potential linkages between $\Delta toIC$'s phenotype and these other non-essential *E. coli* genes, specifically, efflux and lipopolysaccharide genes.

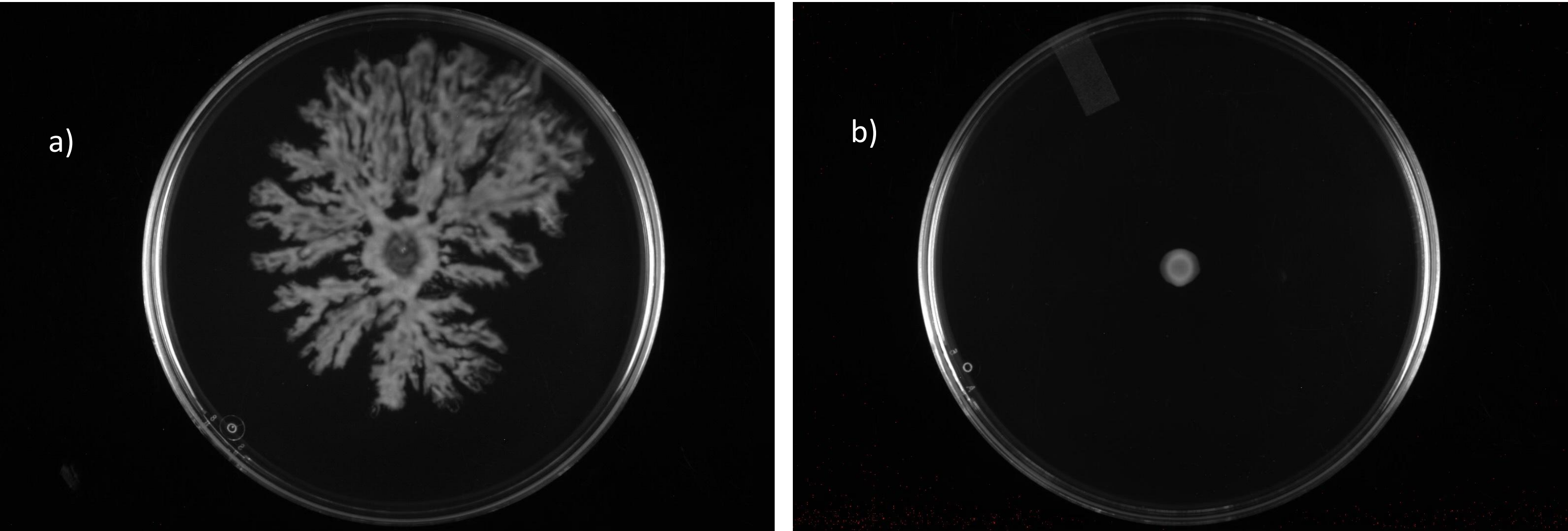


Figure 1. Swarming phenotype of WT CFT073 on semi-solid agar plate (a) and non swarming phenotype of $\Delta toIC$ CFT073 on semi-solid agar plate (b).

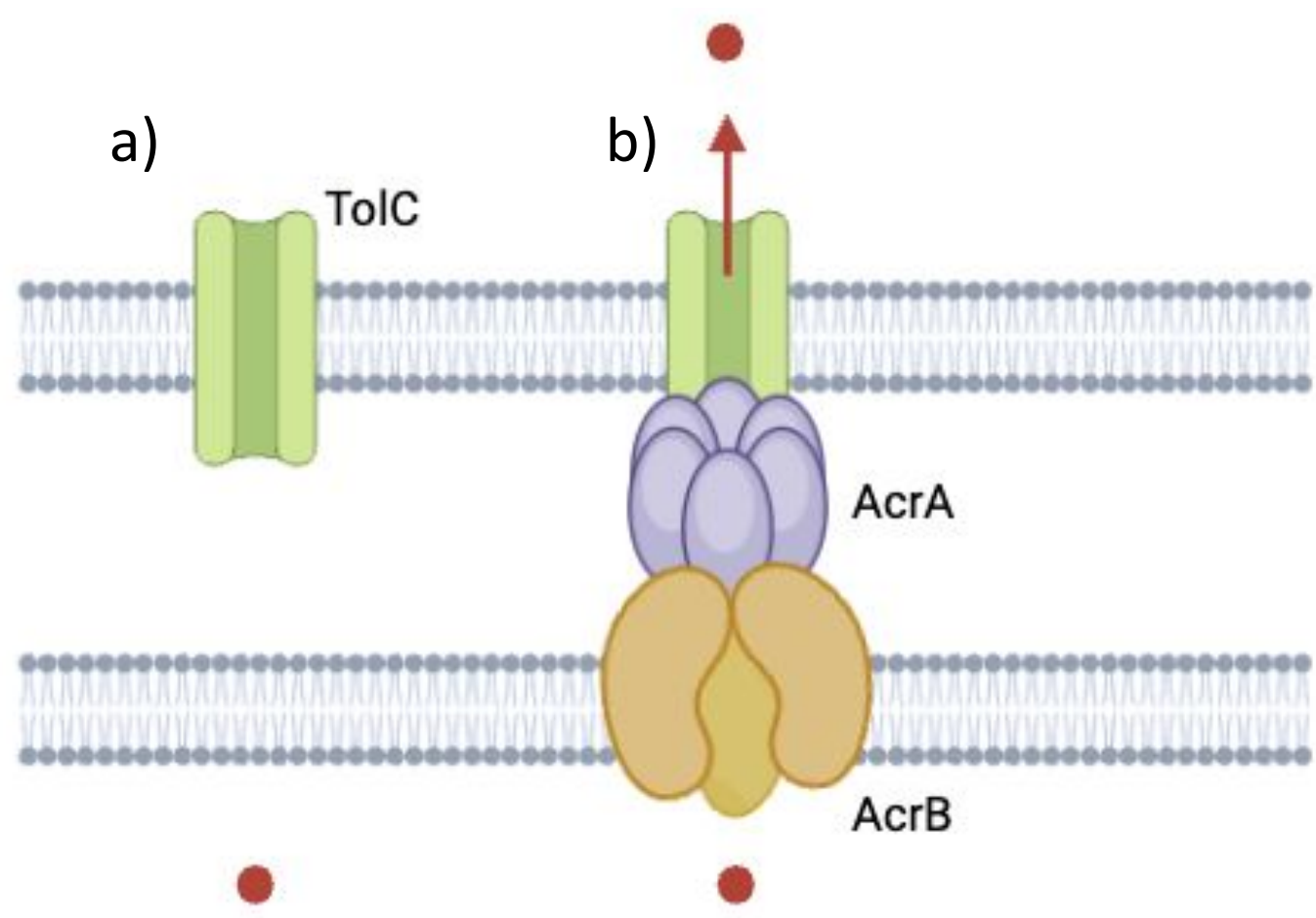


Figure 2. TolC efflux pump embedded in the outer membrane of gram negative bacteria (a). AcrAB TolC complex facilitates the efflux of molecules out of the cell (b). Figure created with BioRender.com.

Results

After investigating 68 genes with unknown function, we found potential linkages between the positive swimming/negative swarming phenotype and genes that code for efflux and lipopolysaccharides. We suggest the following functions for each gene:

- *macA* (Fig. 4c): Involved in the formation of macrolide-specific efflux proteins..
- *mdtH* (Fig. 4d): Involved in the resistance to norfloxacin and enoxacin as a part of the major facilitator superfamily.
- *waaZ*, *wecB*, *wecC*, *wecD*, and *wecF*(Fig. 4e): Involved in biosynthesis of the Enterobacterial common antigens which are OM glycolipids.
- LPS core genes (Fig. 4g): Responsible for the formation of the core component of lipopolysaccharides on the OM.

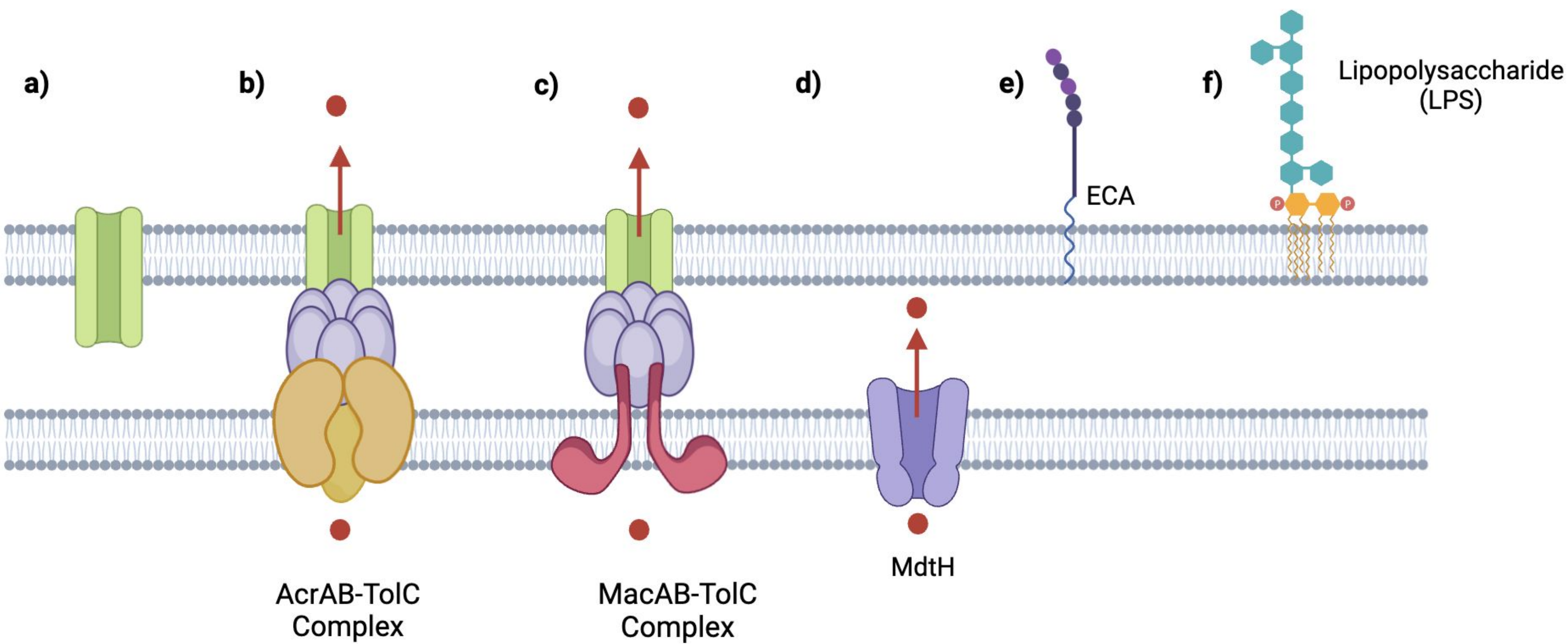


Figure 4. Outer membrane complexes coded by genes proposed to be linked to *E. coli* CFT073 $\Delta toIC$'s phenotype. The TolC efflux pump (a) is recruited by Acr A and Acr B IM proteins to form the AcrAB-TolC complex (b), which effluxes various substrates out of the cell. The MacAB-TolC efflux complex (c) is responsible for effluxing macrolides. mdtH is an IM protein part of the major facilitator superfamily(d). The enterobacteriales common antigen (e) is a carbohydrate antigen conserved through the Enterobacteriales order. Lipopolysaccharides (f) are OM glycolipids that line the surface of gram-negative cells. The structure of lipopolysaccharides play an important role in antibiotic resistance and the formation of other OM structures. Figure created with BioRender.com.

Materials and Methods

- Identified 68 single knockout strains originating from “Genome-Wide Screening of Genes Required for Swarming Motility in *Escherichia coli* K-12”(Inoue et al., 2007) that display a positive swimming phenotype while also displaying a negative swarming phenotype. The 68 genes had no known function at the time of publication.
- Researched and organized the functions of the 68 genes into a gene collection spreadsheet organized by chromosomal location (Fig. 3).
- Focused on genes involving surfactants and LPS.

gene	function	potential link to nonswarming phenotype	references
<i>macA</i>	The membrane fusion protein for the MacAB-TolC tripartite multi-drug transporter. MacA is part of this macrolide-specific efflux protein, primarily effluxing macrolides, but could also be effluxing other compounds given that it confers MDR.	Some classes of Macrolides can be considered as amphipathic, and many surfactants also are (1). Given that the MacA is a part of a multi-drug resistance efflux protein, the substrates it can transport may be diverse (2). These diverse compounds may involve potential surfactants.	https://pubmed.ncbi.nlm.nih.gov/32693319/#:~:text=Surfactants%20are%20normally%20amphiphilic%20molecules,as%20anionic%2C%20cationic%20or%20nonionic,(1)https://pubmed.ncbi.nlm.nih.gov/11544226/ (2)
<i>mdtH</i>	MdtH confers resistance to norfloxacin and enoxacin. MdtH-like MDR transporters belong to the Major Facilitator Superfamily (MFS) of membrane transport proteins, which are thought to function through a single substrate binding site, alternating-access mechanism involving a rocker-switch type of movement.	Norfloxacin and enoxacin could potentially have a similar structure to surfactants or signaling molecules required for the initiation of swarming motility. Further analysis of the compounds are required.(1)	https://www.ebi.ac.uk/interpro/entry/cdd/CD17329/ (1)
ECA gene: <i>wecB</i> , <i>wecC</i> , <i>wecD</i> , <i>wecF</i>	Enterobacterial common antigens are outer membrane surface glycolipids that act as a selective barrier against the permeation of toxic molecules.	Given its amphipathic structure, there is a possibility for surface antigens to act as wetting agents (1). Wetting agents are responsible for decreasing cohesive forces and increasing adhesive forces in water. This would better allow <i>Escherichia coli</i> for swarm.	https://par.nsf.gov/servlets/purl/10170486 (1)
core LPS genes: <i>gmhD</i> , <i>waaF</i> , <i>waaC</i> , <i>waaL</i> , <i>waaU</i> , <i>waaZ</i> , <i>waaY</i> , <i>waaR</i> , <i>waaO</i> , <i>waaB</i> , <i>waaS</i> , <i>waaP</i> , <i>waaG</i> , <i>waaQ</i>	Responsible for the formation of the core component of lipopolysaccharides on the outer membrane.	The integrity of lipopolysaccharides on the outer membrane is important for maintaining normal function of porin and phenotypes [such as swarming] (1). In <i>E. coli</i> mutants that lack heptose region of the inner core, an outer membrane instability phenotype is observed (known as the deep-rough phenotype). "Deep-rough mutants also typically release significant amounts of periplasmic enzymes [degrade compounds in the medium to a utilizable form] into the culture media unless high concentrations of Mg ²⁺ are included in the medium;" (2). Could be possible that the release of periplasmic enzymes saves the swarming phenotype.	https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8448134/ (1) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2569852/ (2) https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2569852/ (3)

Figure 3. Thorough literature review was conducted in order for gene function to be linked to its effect on swarming. The analysis above consists of the genes suspected to play a part in $\Delta toIC$'s non-swarming phenotype.

Discussion/Future Directions

- TolC could be effluxing signaling molecules related to quorum sensing. Could also be effluxing molecules that act as surfactants. We are exploring possible connections between the genes below and the $\Delta toIC$ phenotype.
- MacA confers resistance to macrolide antibiotics, suggesting that MacA could potentially efflux molecules with related chemical structures. These molecules could potentially act as wetting agents/surfactants though further chemical analysis is necessary.
- MdtH confers resistance to norfloxacin and enoxacin. MdtH-like MDR transporters belong to the Major Facilitator Superfamily (MFS) of membrane transport proteins, which are thought to function through a single substrate binding site, alternating-access mechanism involving a rocker-switch type of movement. Structurally similar molecules to norfloxacin and enoxacin could act as surfactants.
- We will generate *macA* and *mdtH* chromosomal deletion mutants in *E. coli* CFT073 to further test the relationship between efflux and swarming.
- Wec proteins are involved within the biosynthesis of Enterobacterial common antigens, which are glycolipids that, along with lipopolysaccharides, make the outer membrane of Enterobacteria. Since ECAs are amphipathic, they may act as potential wetting agents, increasing swarming. We hypothesize that without genes that participate within the biosynthesis of ECAs, swarming for *E. coli* would prove to be difficult.
- Core LPS mutants that lack heptose regions may release periplasmic enzymes that degrade the surrounding substrate. Further investigation into core LPS genes that code for heptose regions should be conducted to understand how LPS genes may influence the swarming phenotype.

References and Acknowledgments

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Inoue, T., Shingaki, R., Hirose, S., Waki, K., Mori, H., & Fukui, K. (2007). Genome-Wide Screening of Genes Required for Swarming Motility in *Escherichia coli* K-12. *Journal of Bacteriology*, 189(3), 950-957.
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